Experiment 1: Decision-trained volatility models on the Moreira–Muir factors

Sep 28, 2026 · @Rob

The GARCH-LSTM network on its own still adds nothing to the portfolio. Trained for the decision, it now helps robustly on SMB and HML, and in the same direction on RMW and CMA: on those four factors every one of five objective-B seeds beats every objective-A seed. It hurts on Mkt.

Against a plain GJR, only the SMB gain (+0.20 Sharpe) survives every multiple-testing correction. Outside Mom, no version passes Cederburg's real-time test. The first round, with our own simpler GARCH-LSTM and one or two seeds, is kept below the update for reference.

Update: five seeds, Zhao et al.'s architecture, corrected tests

Three changes answer the weaknesses of the first round.

- **Zhao et al.'s GARCH-LSTM** replaces our overlay. It follows their equation (7): LSTM gates fed yesterday's shock and variance, the GJR kernel as output gate, and $\sigma^2 = o \cdot (1 + w \cdot \tanh(c))$, with their 126-day input window and ×100 scaling. The network now sits inside the GARCH recursion.
- **Five seeds per annual refit**, for objective A and objective B, with the five portfolio weights averaged.
- **36 tests fixed in advance**, on Sharpe ratios net of costs, drawn from one joint block bootstrap and adjusted three ways: Holm, Benjamini–Hochberg and Romano–Wolf. Forecast accuracy is tested separately with Diebold–Mariano on monthly QLIKE.

Factor	Buy-and-hold	MM rule	GJR · A	Zhao · A	Zhao · B	Zhao · B, range over 5 seeds
Mkt	0.52	0.41	0.45	**0.45**	0.39	0.30–0.42
SMB	0.17	0.05	0.08	0.09	**0.28**	0.24–0.30
HML	0.39	0.26	0.38	0.37	**0.47**	0.42–0.49
Mom	0.50	0.65	0.85	0.90	**0.94**	0.88–0.96
RMW	0.41	0.54	0.52	0.53	**0.65**	0.63–0.67
CMA	0.44	0.30	0.37	0.37	**0.44**	0.41–0.44

Net Sharpe ratio, out-of-sample 1937–2026 (RMW, CMA 1974–2026); bold = best managed version. Objective A is very stable across seeds (range at most 0.03), objective B much less so (up to 0.13 on Mkt), which is why the ensemble matters.

Factor	Network alone: Zhao A − GJR A	Decision training: Zhao B − Zhao A	Full package: Zhao B − GJR A
Mkt	+0.01 (0.70 / 1.00)	−0.06 (0.31 / 0.99)	−0.05 (0.34 / 1.00)
SMB	+0.02 (0.11 / 0.86)	**+0.19 (0.003 / 0.08)**	**+0.20 (0.001 / 0.03)**
HML	−0.01 (0.07 / 0.76)	**+0.10 (0.008 / 0.16)**	+0.10 (0.014 / 0.26)
Mom	**+0.05 (0.004 / 0.10)**	+0.04 (0.39 / 1.00)	+0.09 (0.10 / 0.82)
RMW	+0.01 (0.42 / 1.00)	+0.12 (0.08 / 0.78)	+0.14 (0.04 / 0.54)
CMA	+0.00 (0.75 / 1.00)	+0.06 (0.27 / 0.99)	+0.07 (0.26 / 0.99)

Net Sharpe difference (raw p / Romano–Wolf p over all 36 tests). Bold = still significant at 5% under Benjamini–Hochberg; the SMB full-package gain also survives Holm (0.04) and Romano–Wolf (0.03). For GARCH and GJR, objective B still never helps (all 12 tests $p > 0.15$).

Forecast accuracy goes the other way. Zhao's model trained on A is more accurate than GJR on Mkt and Mom (Diebold–Mariano $p = 0.03$ and < 0.001), less accurate on HML and RMW ($p = 0.002$ and 0.03), and equal on SMB and CMA. Trained on B, its forecasts are far worse on five factors: QLIKE loss 2 to 4.5 times higher, all $p < 0.001$. So the gains come from reshaping the signal, not from predicting variance better.

The Cederburg test still fails outside Mom. Combining Zhao · B with the unmanaged factor in real time adds +0.07 to +0.16 Sharpe on SMB, HML and RMW, but none of it is significant ($p = 0.29$–0.51); Mom gains +0.55 ($p < 0.001$). After publication (2016–2026), B minus A ranges from −0.08 (Mkt) to +0.18 (RMW, $p = 0.09$), with nothing significant.

What we had to choose because Zhao et al. leave it open: w is an 8-dimensional vector with $\Sigma|w| < 1$ (keeps variance positive), the loss is Gaussian QMLE rather than their Student-t (so objective A is the same for all models), each 126-day window starts from the GJR-filtered variance, and the month-ahead forecast uses GJR mean reversion from the model's one-day forecast.

Setup

We rerun the Moreira & Muir (2017) experiment on their own universe and benchmark, and swap only how the variance forecast v is made. Each version is then run in real time, the way Cederburg et al. (2020) test the rule.

Choice	What we did
Data	Ken French data library, CRSP vintage Aug 2026, daily and monthly
Universe	Mkt-RF, SMB, HML, Mom, RMW, CMA (the six French factors in MM Table I)
Benchmark	The unmanaged factor, buy-and-hold
Trading rule	Monthly weight $w = c / v$, managed return = $w \times$ next month's factor return, no leverage cap
Scaling c	Set each January on past months only, so the managed series has the unmanaged volatility in-sample
Forecasts	MM rule (last month's realised variance); GARCH(1,1), GJR-GARCH and GARCH-LSTM fitted on daily returns, 21-day-ahead variance
Objective A	Daily Gaussian quasi-likelihood (QMLE)
Objective B	Monthly utility of the managed portfolio, mean(R) − (γ/2) mean(R^2) with γ = 5 and 10 bp per unit of weight change; starts from the A solution, early-stops on the last 20% of training months, mean held at its A value
Real-time protocol	Expanding window, all models refit every January on data before that date; first trade after 10 years of data (1937 for Mkt, SMB, HML, Mom; 1974 for RMW, CMA); last month Aug 2026
Periods reported	Full out-of-sample; 1937–2015 (MM's sample end); 2016–2026 (after publication)

The GARCH-LSTM is our own version of the Zhao et al. (2024) idea: a GJR-GARCH variance multiplied by exp(g), where g is the output of a small LSTM (8 units) reading the last 63 trading days of standardised shocks and log variance. The output layer starts at zero, so the network starts exactly at its GJR solution and only learns a correction.

Replication check

With full-sample c, as in the paper, we reproduce MM Table I Panel A: the managed market earns 4.88% a year of alpha (s.e. 1.56) against their 4.86% (1.56). Remaining gaps come from French's later data revisions and our sample starting one month after theirs (1,073 months vs their 1,065).

Factor	Alpha, ours (% p.a.)	Alpha, MM (% p.a.)	Beta, ours	Beta, MM	Months
Mkt	4.88 (1.56)	4.86 (1.56)	0.60	0.61	1,073
SMB	−0.58 (0.91)	−0.58 (0.91)	0.61	0.62	1,073
HML	1.65 (1.04)	1.97 (1.02)	0.54	0.57	1,073
Mom	12.80 (1.71)	12.51 (1.71)	0.45	0.47	1,068
RMW	2.81 (0.88)	2.44 (0.83)	0.58	0.62	629
CMA	0.28 (0.67)	0.38 (0.67)	0.67	0.68	629

Heteroskedasticity-robust standard errors in parentheses; 1926–2015 (RMW and CMA from 1963).

Out-of-sample Sharpe ratios

In real time, the MM rule beats buy-and-hold only on Mom and RMW. In this run, the GARCH-LSTM trained for the decision (LSTM · B) is the best managed version on five of six factors after costs, and the only one that beats buy-and-hold on SMB, HML and CMA. Part of that lead depends on the network's random seed (see Robustness).

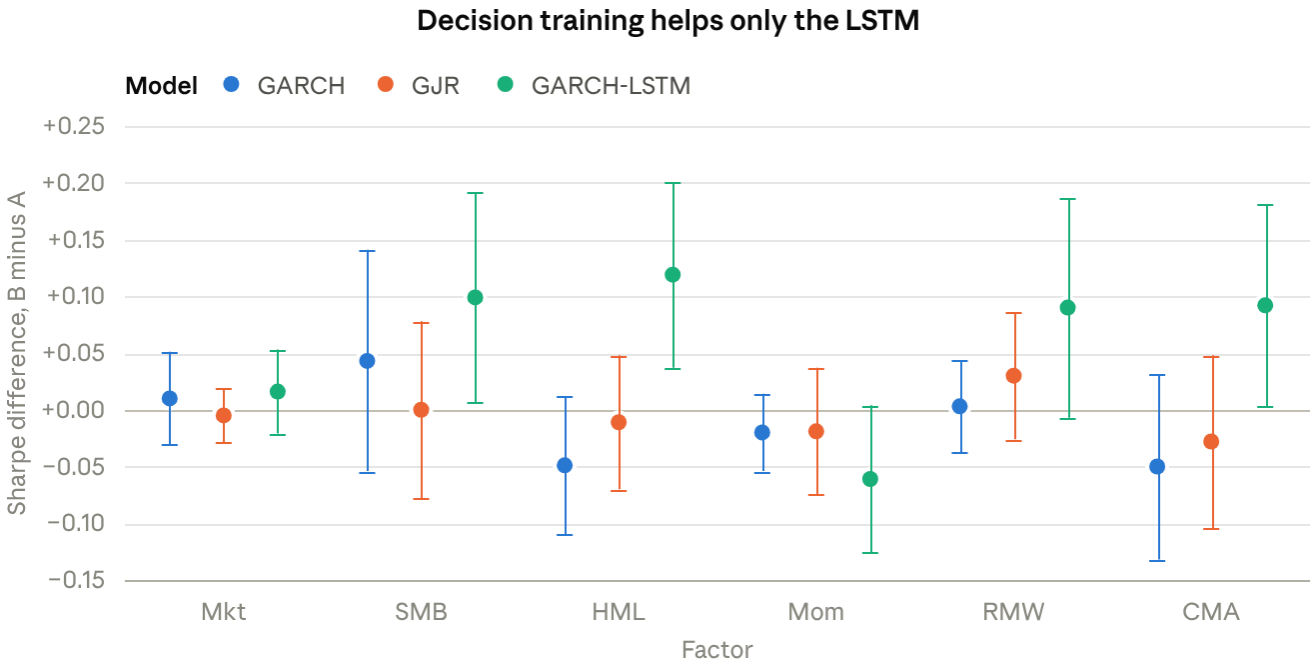
Factor	Buy-and-hold	MM rule	GARCH · A	GARCH · B	GJR · A	GJR · B	LSTM · A	LSTM · B
Mkt	0.52	0.41	0.44	0.45	0.45	0.44	0.46	**0.47**
SMB	0.17	0.04	0.04	0.11	0.08	0.09	0.11	**0.19**
HML	0.39	0.26	0.37	0.34	0.38	0.39	0.35	**0.46**
Mom	0.50	0.65	0.85	0.82	0.85	0.83	**0.86**	0.79
RMW	0.41	0.54	0.50	0.50	0.52	0.54	0.48	**0.55**
CMA	0.44	0.30	0.39	0.37	0.37	0.37	0.38	**0.46**

Annualised Sharpe ratio net of 10 bp per unit of weight change; bold = best managed version. Out-of-sample 1937–2026 (1,076 months), RMW and CMA 1974–2026 (632 months). Before costs the ranking is the same except RMW, where the MM rule ties LSTM · B at 0.63.

Better forecasts also halve trading. The MM rule changes its weight by 0.35–1.29 per month on average, while the GARCH family changes it by 0.10–0.74. The MM rule's 99th-percentile weight is 3–13×, against 2–5× for the models.

Objective B vs objective A

Training for the decision raises the GARCH-LSTM's Sharpe ratio by 0.09–0.12 on SMB, HML, RMW and CMA ($p = 0.004$–0.07). It leaves Mkt unchanged and costs 0.06 on Mom ($p = 0.07$). For GARCH and GJR, B is never better than A.



Sharpe ratio of objective B minus objective A, same model, full out-of-sample. Dot = difference, whisker = 95% block-bootstrap interval (12-month blocks).

results/ab_tests.csv · 6 factors × 3 models · 1937–2026

The pattern fits the proposal's hypothesis: decision training pays when the model has enough freedom to reshape its signal. GARCH has three variance parameters, and B mostly lowers α and raises β at the same persistence, giving a smoother forecast. B's forecasts are also *less accurate*: the GARCH-LSTM's QLIKE loss rises on every factor, and its log-forecast correlation with realised variance drops by 0.02–0.14. That is the Kim et al. (2025) pattern of better portfolios from worse forecasts. These are 18 tests from one random seed. A second seed (Robustness) keeps the direction but shrinks the gains, except on RMW.

Better forecasts, and the Cederburg test

Replacing last month's realised variance with a QMLE-fitted GJR forecast raises the Sharpe ratio on five of six factors. The gain is significant on HML (+0.09, $p = 0.01$) and Mom (+0.18, $p = 0.05$), and it halves turnover. RMW is the exception, where the MM rule is

better (−0.07, p = 0.07).

Cederburg et al.'s sharper test asks whether an investor who combines the managed and unmanaged factor in real time, with mean-variance weights estimated only on past months, beats holding the factor alone. Only Mom passes, for every version of v. LSTM · B is the only version that never lowers the combination's Sharpe ratio, but none of its gains outside Mom is significant.

Factor	Buy-and-hold Sharpe	MM rule	GJR · A	LSTM · A	LSTM · B	Test starts
Mkt	0.55	+0.02 (0.51)	−0.00 (0.92)	+0.02 (0.51)	+0.03 (0.38)	1947
SMB	0.11	−0.04 (0.04)	−0.04 (0.14)	−0.06 (0.02)	−0.00 (0.97)	1947
HML	0.37	−0.02 (0.42)	+0.00 (0.89)	−0.01 (0.77)	+0.09 (0.31)	1947
Mom	0.58	+0.29 (0.00)	+0.48 (0.00)	+0.47 (0.00)	+0.46 (0.00)	1947
RMW	0.48	+0.09 (0.64)	+0.04 (0.83)	−0.01 (0.94)	+0.04 (0.81)	1984
CMA	0.32	−0.05 (0.28)	−0.06 (0.22)	−0.06 (0.23)	+0.01 (0.90)	1984

Sharpe ratio of the real-time combination minus buy-and-hold, bootstrap p-value in parentheses. Weights use at least 120 past out-of-sample months, so the test starts 10 years after trading does.

After volatility spikes

LSTM · B gets back in faster after a spike than the MM rule, and on four of six factors it was right to. In the six months after a spike it holds 38–78% of its usual weight, against 30–64% for the MM rule. Its post-spike Sharpe ratio is higher everywhere except HML and Mom. On Mom, staying out, as the MM rule does, was the better call.

Factor	Spikes	Buy-and-hold Sharpe after	MM rule	GJR · A	LSTM · A	LSTM · B
Mkt	53	1.22	0.48 · 0.67	0.57 · 0.98	0.59 · 1.02	0.62 · 1.10
SMB	58	0.34	0.64 · −0.16	0.60 · −0.02	0.64 · 0.06	0.78 · 0.29
HML	70	−0.08	0.36 · 0.09	0.41 · 0.13	0.44 · 0.03	0.43 · −0.05
Mom	62	0.59	0.30 · 0.84	0.35 · 0.66	0.37 · 0.66	0.38 · 0.68
RMW	66	0.37	0.47 · 0.54	0.47 · 0.54	0.56 · 0.58	0.64 · 0.68
CMA	57	0.25	0.56 · 0.07	0.57 · 0.15	0.61 · 0.08	0.73 · 0.27

Each cell: average weight in the 6 months after a spike, relative to the strategy's median weight · annualised Sharpe ratio over those months. A spike is a month whose realised variance tops the 95th percentile of all months seen so far (real-time definition). Post-spike months overlap across spikes, so these are descriptive, not tests.

Robustness

A second random seed for the GARCH-LSTM keeps the sign of B minus A on every factor except Mkt, but shrinks the gain on SMB, HML and CMA to about +0.02. RMW is the one factor where the gain holds at full size under both seeds. Averaging the two seeds' weights (a two-network ensemble), B beats A by +0.10 on RMW and +0.08 on HML, both p = 0.006.

Factor	Seed 1: B − A	Seed 2: B − A	Two-seed ensemble: B − A	Ensemble Sharpe, A → B	Buy-and-hold Sharpe
Mkt	+0.02 (0.40)	−0.01 (0.62)	+0.01 (0.78)	0.47 → 0.48	0.52
SMB	+0.10 (0.04)	+0.01 (0.80)	+0.06 (0.11)	0.13 → 0.19	0.17
HML	+0.12 (0.00)	+0.02 (0.43)	+0.08 (0.01)	0.38 → 0.46	0.39
Mom	−0.06 (0.07)	−0.02 (0.60)	−0.03 (0.07)	0.90 → 0.87	0.50
RMW	+0.09 (0.07)	+0.09 (0.04)	+0.10 (0.01)	0.55 → 0.65	0.41
CMA	+0.09 (0.04)	+0.02 (0.71)	+0.07 (0.10)	0.43 → 0.49	0.44

Gross Sharpe ratios, full out-of-sample; bootstrap p-value in parentheses. Seed 1 is the run behind every table above.

After publication (2016 to Aug 2026, 128 months) nothing is significant. B minus A for the GARCH-LSTM ranges from −0.11 (CMA) to +0.12 (RMW). Every managed version of Mkt trails buy-and-hold (0.84), which is the Cederburg result again, in a strong bull market.

Caveats and next steps

- **Seed sensitivity is the main weakness.** Objective B picks its stopping point on 20% of the training months, which is a noisy signal. Done in the update: 5 seeds per refit, ensemble reported.
- **Multiple testing.** 18 B-vs-A comparisons at 5% would give about one false positive by chance. The seed 1 results show four; the update corrects for this.
- **Our GARCH-LSTM is not Zhao et al.'s exact architecture.** It is a GJR model with a bounded LSTM correction (at most ×e either way). Done in the update: their equation (7) is now implemented.

- **Objective B uses vol-matched c inside training.** That makes B close to "maximise the net Sharpe ratio", so γ barely matters. A version with c free and a leverage cap of 1.5 (MM Table V) would test the utility itself.
- **The factors are long-short portfolios,** and trading costs for SMB, HML, RMW and CMA are rougher than for Mkt. MM only apply costs to Mkt.
- **Next experiment:** the same six versions on the course's equal-weighted US stock portfolio, 2000–2020, using the LSEG daily panel.

Code, results and the downloaded French files are in the experiment folder. Run `backtest.py` then `evaluate.py` to reproduce everything; the full run takes about 1.5 hours on 2 cores.